

A Carbon Futures Contract for the Guangzhou Futures Exchange

I. Underlying Asset and Trading Product

This design takes covered emitters’ allowance procurement risk as its starting point and selects China Emission Allowances (CEA) as the underlying asset for carbon futures on the Guangzhou Futures Exchange (GFEX). Firms must surrender allowances against verified emissions[21]; when available allowances fall short, they must buy the balance, so future allowance prices directly affect compliance costs. CEA futures offer a way to manage this procurement risk in advance. In 2025, China’s national carbon market covered 3,378 emitters in power generation, steel, cement, and aluminium smelting[10], providing a broad base of potential hedgers. A futures underlying also needs standardised asset units, regular spot trading, and enough deliverable supply to support price formation and convergence and reduce manipulation risk. The following sections compare CEA and China Certified Emission Reductions (CCER) against these criteria.

A. Suitability for Standardisation

CEA has fewer asset differences to address and an existing, consistently published market price reference. CEA eligibility mainly depends on vintage, registry status, compliance validity, and transferability[5, 7]; CCER also differs by project origin, methodology, issuance year, and registration batch[12]. The national registry identifies CEA ownership and status under common rules[5], making it easier to define delivery standards by vintage and availability.

The Shanghai Environment and Energy Exchange regularly publishes composite CEA prices[19], providing a common benchmark for linking spot and futures prices. CCER has **price differences across projects**: in 2025, annual average transaction prices for grid-connected solar thermal and offshore wind projects were CNY 81.58 and 69.27 per tonne, respectively[17]; averages across projects also vary with the transaction mix.

B. Price Volatility

CEA price changes directly affect compliance procurement costs for firms with allowance shortfalls, creating a clear hedging need. In 2025, composite CEA prices ranged from CNY 50.34 to 97.01 per tonne, and the year-end closing price fell 23.45% from the previous year-end[18]; over the common trading window, the standard deviation of daily price changes was CNY 1.07 per tonne. Procurement costs therefore face observable, measurable price risk.

Figure 1(a) shows frequent sharp CCER price moves over short periods, which raise daily mark-to-market gains and losses and margin requirements, increasing hedgers’ funding costs. Excessive volatility may also weaken the stability of the spot–futures price relationship and reduce hedge effectiveness. Figure 1(b)–(c) shows highly concentrated CCER trading: the ten largest trading days account for 65.40% of volume. This suggests limited continuity in routine CCER trading and greater manipulation risk.

Table 1. Trading Base, Activity, and Liquidity of CEA and CCER

Indicator	CEA	CCER
<i>A. Trading base and participants</i>		
Supply and demand base	Annual allocation; mandatory surrender[21]	Project verification and registration; 5% compliance offset[12]
Actual trading participants	1,471 (2024)[8]	90 (cumulative to August 2025)[8]
<i>B. Listed trading in the common window</i>		
Volume (million tonnes)	79.0117	8.8441
Turnover (CNY billion)	4.949	0.626
Median daily volume (tonnes)	288,406	2,700
Days with volume $\leq 1,000$ tonnes	5	72
Ten largest days’ share (%)	16.35	65.40
Fourth-quarter volume share (%)	58.93	68.28
<i>C. Standardised liquidity measures</i>		
Amihud ^a	0.001197	13.059870
Standard deviation of daily price changes (CNY/tonne) ^b	1.07	7.14

Notes: The sample covers 203 common trading days from 7 March to 31 December 2025 and uses listed trades; daily volume measures retain zero-trade days. ^aAmihud is the sample mean of $|r_t|/(A_t/10^6)$ [1], where r_t is the simple return in decimal form and A_t is daily turnover in CNY; price pairs are retained only for consecutive trading days with listed trades on both days. CEA uses the composite closing price and CCER the daily average across projects; these different price definitions are used here only to assess CEA’s suitability as a futures underlying. ^bDaily price changes are differences between consecutive trading days, retaining only pairs with listed trades on both days. *Sources:* CEA trading and price data[19]; CCER totals and turnover[13, 16, 17]; CCER daily trading and price data[2], corrected against official notices[14, 15]. Panel B and C statistics are calculated here.

C. Potential Hedging Demand

Potential demand for long CEA hedges comes from allowances that firms still need to buy before compliance. In the first compliance cycle (2019–2020), 847 firms had allowance shortfalls, representing 39.18% of covered firms, with a combined shortfall of about 188 million tonnes[6]. Recent corporate disclosures further illustrate the scale of procurement exposure: after deducting its allowance balance, Changyuan Power estimated a 2025 compliance shortfall of about 511,200 tonnes and approved the corresponding purchases in 2026[3]; at that volume, a CNY 10 per tonne price rise adds about CNY 5.112 million to procurement costs. Firms with shortfalls thus have an economic need to lock in CEA purchase prices in advance.

Managing compliance procurement prices is also seasonal. The fourth quarter accounted for 79% of annual CEA volume in 2024[8] and 58.18% in 2025[19] (Figure 1(d)). This concentration of trading near year-end encourages firms to manage procurement risk before compliance.

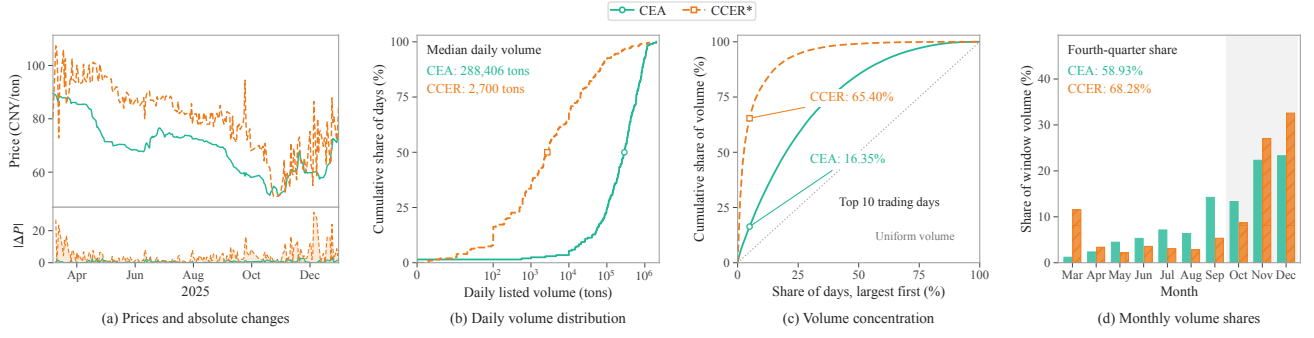


Figure 1. CEA and CCER Price Changes and Listed Trading Distributions

Notes: The sample covers 203 common trading days from 7 March to 31 December 2025 and retains zero-trade days. The secondary CCER daily table is 9,020 tonnes below the official cumulative total; daily distributions use the compiled sample. (a) The upper panel shows the CEA composite closing price and CCER daily average transaction price; the lower panel shows absolute price differences between adjacent observation days, $|P_t - P_{t-1}|$, both in CNY per tonne. (b) The empirical cumulative distribution of daily volume uses a logarithmic horizontal axis, with ticks in tonnes. (c) Cumulative volume shares rank days from highest to lowest volume and mark the ten largest days; the grey diagonal represents equal volume on all days. (d) Monthly volume is expressed as a share of each market's sample total; the grey background marks the fourth quarter. *Sources:* Shanghai Environment and Energy Exchange daily notices[19]; daily tables compiled by Carbon Neutrality Network[2], corrected against national CCER trading system notices[14, 15]. Calculations by the author.

D. Underlying Asset and Deliverable Allowances

The benchmark deliverable is CEA for the latest compliance vintage: allowances for the emissions year whose surrender is due in the delivery year, with the vintage fixed at listing. For example, compliance in 2026 covers 2025 emissions, making 2025-vintage CEA the benchmark[11]. Substitutes are limited to other CEA vintages that the Ministry of Ecology and Environment (MEE) recognises as equally valid for that compliance year and that have completed any required carryover registration; current rules allow 2026-vintage allowances to meet 2025 surrender obligations[9]. Allowances from 2019–2024 are eligible only after being withdrawn and reissued as 2025-vintage allowances under the carryover rules; unconverted old allowances are ineligible[7, 9].

Deliverable allowances must be registered in the seller's own national registry account and available for normal use; both parties must be eligible to hold and transfer or receive spot allowances and to make or take delivery on the exchange[5]. This proposal excludes surrendered, cancelled, frozen, pledged, or otherwise transfer-restricted allowances, as well as allowances locked by pending surrender, cancellation, or carryover applications; eligibility must be rechecked once the restriction ends. Benchmark and eligible substitute allowances may be mixed at one tonne for one tonne, but must be declared and checked by vintage to ensure that all are valid for the same compliance year. If policy changes compliance validity or carryover conditions, the exchange should update its eligibility list and require replacement of invalid allowances; if no valid substitute exists, it should suspend affected deliveries and resolve outstanding contracts under published contingency rules.

II. Standardised Contract Specifications

The trading unit is 1,000 tonnes of CO₂e per contract. ICE Exend EUA futures use 1,000 allowances per contract, each representing one tonne of CO₂e[4]; domestic purchases have reached 100,000 tonnes, including Sinopec's CEA purchase from China Resources Group on 21 July 2021[20], which

Table 2. Proposed Carbon Futures Contract Specifications

Term	Design content	Design basis
Trading product and code	[To complete: Proposed official name and code]	[To complete: Underlying definition, code uniqueness check, and source]
Trading unit	1,000 tonnes of CO ₂ e per contract	Based on EUA futures and a domestic purchase of 100,000 tonnes[4, 20]; this equals 100 contracts, with rounding to the nearest whole contract giving a maximum mismatch of 500 tonnes. At the 2025 year-end price of CNY 74.63 per tonne[10], contract value is CNY 74,630.
Quotation unit	CNY per tonne of CO ₂ e	Matches the CEA unit[21] and spot quotation[19]; profit or loss per contract equals the price change multiplied by 1,000 tonnes.
Minimum price fluctuation	[To complete: Amount per quotation unit]	[To complete: Quotation precision; tick value = tick size × trading unit]
Maximum daily price fluctuation limit	[To complete: Percentage of prior settlement price and exceptional rules]	[To complete: Daily volatility, compliance-season tail risk, and stress tests; do not copy the illustrative ±6%]

Table 2. Contract Specifications (Continued)

Term	Design content	Design basis
Minimum trading margin	[To complete: Rate and escalation conditions]	[To complete: Contract value $\times$ rate; risk coverage and compliance-season adjustments]
Contract months	[To complete: Months and listing horizon]	[To complete: Hedging horizon, allocation and surrender calendar, and liquidity]
Last trading day	[To complete: Operational trading-day rule]	[To complete: Time for price publication or asset transfer and holiday treatment]
Delivery method	[To complete: Physical delivery or cash settlement]	[To complete: Account and system feasibility or spot-index feasibility; match Section III]
Delivery grade/standard	[To complete: Eligible assets or cash-index sample]	[To complete: Vintage or batch, registry status, exclusions, and consistency with the price source]

would equal 100 contracts under this proposal. Rounding to whole contracts gives a maximum volume mismatch of 500 tonnes, or just 0.5% of a 100,000-tonne exposure. This size balances hedge granularity against contract value; specifications appear in Section II.

III. Delivery and Reference and Settlement Prices

Method and rationale: [To complete: Compare physical delivery and cash settlement; state the choice and prerequisites]. For physical delivery, specify the seller's eligible assets and accounts, freezing or transfer, payment, delivery failure procedures, and the verification status of the registry interface. For cash settlement, specify the index sample, trading method, data licence, observation window, weighting formula, and manipulation safeguards.

Daily settlement price: [To complete: Futures transaction data source, calculation window and formula, publication time, and rules for no trades or anomalies].

Final price at expiry: [To complete: Data source, formula, observation window, publication time, correction and fallback rules for the delivery settlement price or final cash settlement price]. Explain how this price converges with the underlying spot price; keep the spot closing price, daily futures settlement price, and expiry price distinct.

IV. Risk Management

Margin and daily mark-to-market: [To complete: Base rate, escalation conditions near expiry or during exceptional volatility, and calculation basis].

Price limits: [To complete: Reference price, limit width, and measures after consecutive limit moves].

Position limits: [To complete: Limits for ordinary months and near expiry, account aggregation, and hedge treatment].

Large-position reporting: [To complete: Trigger, required information, reporting time, and verification procedure].

Compliance cycle: [To complete: Applicable surrender or cancellation dates, potential price and supply shocks, and

triggers for adjusting margin and position limits]. Briefly describe fallbacks for registry or price-source outages, no trades, policy changes to eligible assets, and concentrated positions, consistent with Table 2 and the delivery terms.

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